CHANDERNAGORE COLLEGE

SEMESTER –V, B.A. / B.Sc. / B.COM (MAJOR) INTERNAL ASSESSMENT EXAMINATION, 2025

SUBJECT: COMPUTER SCIENCE

COURSE CODE: COMP 5011

COURSE NAME: Data Communications & Networking

FULL MARKS: 10

TIME: 30 MINUTES

ATTEMPT ANY 02 OUT OF THE FOLLOWING 03 QUESTIONS

2 x 5 = 10

1. Draw the star topology with explanation. Write the advantages and disadvantages of ring topology. 5
2. Draw a neat diagram of the OSI (Open Systems Interconnection) Reference Model. List all seven layers in the correct order and explain the primary function of the Data Link Layer and the Transport Layer. (3+2)
3. a) Differentiate between Guided and Unguided transmission media. Compare Coaxial Cable, Twisted Pair, and Fiber Optic Cable based on bandwidth capacity and EMI (Electromagnetic Interference) resistance.
   b) Explain the construction of a Fiber Optic Cable. Why is it considered superior to copper cables for long-distance communication? (2+3)